

Start with PowerBI & AI

"From Rookie to Rock"

Learn & Practice
FREE PowerBI pbix

Profitability of a project

Financial UDF pack

Part 2: DAX Code guide



Patou Tips #60



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Learn & Practice
FREE PowerBI pbix

Finance UDF (User Defined Function) Pack

Profitability of a project

Part 2: DAX Code guide

💡 Find the DAX Code on Patou Tips
GitHub, # 60, file READ.md, chapter 3
→ <https://github.com/PatouTips/Patou-Tips/>



Profitability of a project DAX Code guide (1/7)



Find the DAX Code on Patou Tips
GitHub, # 60, on the file
README, chapter 3

<https://github.com/PatouTips/Patou-Tips/>

About this financial package...

How it works?

1 - Parameters

2 - Usage

3 - DAX Code

DF - Discount Factor

FCF - Free Cash Flow

DCF - Discounted Cash Flow

NPV - Net Present Value

NPV Total - Net Present Value

IRR - Internal Rate of Return

Payback (Year)

Payback (number of years)

Financial explanations

License

README

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Discount Factor

Evaluate the value of one euro (or other money) according to a required rate of return and the number of term. Required rate of return is the Discounted Rate and it can be based on the WACC, the inflation rate, the financial internal rate of return...

$$\text{Discount Factor} = 1 / (1 + \text{Discounted Rate})^{(\# \text{Period} - 1)}$$

Year	Discount rate
2024	1,00
2025	0,91
2026	0,83
2027	0,75
2028	0,68
2029	0,62
2030	0,56
2031	0,51
2032	0,47

Example with a Discounted Rate of 10% and a project calculation at 2028:

$$0,68 = 1 / (1 + 10\%)^{((2028 - 2024 + 1) - 1)}$$

Dax measure

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1

Profitability of a project DAX Code guide (3/7)

Free cash flow (FCF) is the money a company has left over after paying its operating expenses and capital expenditures.

Free Cash Flow (FCF) = Revenue + Costs + Investment

Year	Revenue		Cost		Investment		Free Cash Flow (FCF)
2024	150 000 €		-50 000 €		-200 000 €		-100 000 €
2025	160 000 €		-100 000 €		-30 000 €		30 000 €
2026	190 000 €		-150 000 €		-10 000 €		30 000 €
2027	230 000 €	+	-150 000 €	+	-10 000 €	=	70 000 €
2028	260 000 €		-150 000 €		-10 000 €		100 000 €
2029	275 000 €		-150 000 €		-10 000 €		115 000 €
2030	300 000 €		-150 000 €		-10 000 €		140 000 €
2031	330 000 €		-150 000 €		-10 000 €		170 000 €
2032	360 000 €		-150 000 €		-10 000 €		200 000 €

Dax measure

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Discounted cash flow (DCF) is a valuation method that estimates the value of an investment using its **expected future cash flows**.

Discounted Cash Flow (DCF) = Free Cash Flow x Discount Factor

Year	Free Cash Flow (FCF)		Discount rate		Discounted Cash Flow (DCF)
2024	-100 000 €	X	1,00	=	-100 000 €
2025	30 000 €		0,91		27 273 €
2026	30 000 €		0,83		24 793 €
2027	70 000 €		0,75		52 592 €
2028	100 000 €		0,68		68 301 €
2029	115 000 €		0,62		71 406 €
2030	140 000 €		0,56		79 026 €
2031	170 000 €		0,51		87 237 €
2032	200 000 €		0,47		93 301 €

Dax measure

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NPV - Net Present Value

When the NPV is positive, the project is adding value to by generating revenues that exceed the costs, once they are discounted.

NPV = Cumulative Discounted Cash Flow

Year	Discounted Cash Flow (DCF)	NPV
2024	-100 000 €	-100 000 €
2025	27 273 €	-72 727 €
2026	24 793 €	-47 934 €
2027	52 592 €	4 658 €
2028	68 301 €	72 959 €
2029	71 406 €	144 365 €
2030	79 026 €	223 392 €
2031	87 237 €	310 629 €
2032	93 301 €	403 930 €



NPV must be positive

Dax measure

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This project starts creating value from the fourth year, means that this project must have at least 4 years as a term to be profitable

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IRR – Internal Rate of Return

The IRR is the **discount factor** at which the net present value (NPV) of a set of **discounted cash flows (DCF)** is **equal to zero**. On PowerBI or Excel, creating such a calculation is so complex, fortunately there is the "XIRR" function.

Dax measure

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IRR limits:

- The IRR assumes that the project's future cash flows are reinvested at the same rate as the IRR, which may not be realistic or achievable. ...
- Sometimes project A may have a higher IRR than project B while project B generates more cash flow (NPV).



Payback Period

The **payback period** is the **amount of time** it takes to recover the cost of an investment. Simply put, it is the length of time an investment reaches a **breakeven point**.

A payback period relatively short indicates a rapid return on investment, that reduces risk and improves liquidity.

Payback Period = (Project Start Date - Date when NPV >= 0) +1



The payback period must be less than or within the "project duration"

Dax measure

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Coming soon, in 2026 / 2027!



Patrice Fayard

Business Intelligence WORKBOOK

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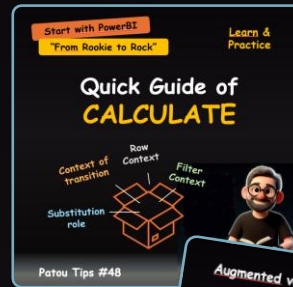


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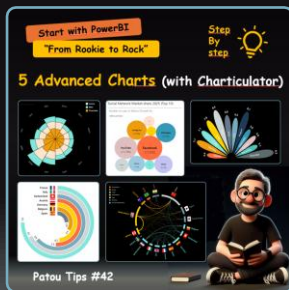
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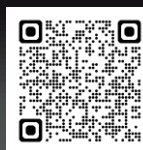
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Resources on GitHub
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Patou Tips



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